

Hand Calculations

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MUSIC group: $n_1 = 48$ $\mu = 4.125$ $\sigma = 0.789$

Silence group: $n_2 = 18$ $\mu = 2.611$ $\sigma = 1.378$

} gathered from survey

Significance level:

$\alpha = 0.05$

Hypotheses:

$H_0: \mu_{\text{music}} = \mu_{\text{silence}}$

$H_a: \mu_{\text{music}} \neq \mu_{\text{silence}}$

Conditions:

10% Condition:

66 students ≤ 0.10 (all Highschool Students in Dorai Academy)

Independence

- Each response is independent, no duplicate submissions.

Randomness

- Survey was distributed randomly.

Two Sample T-Test for the Difference of two means

Calculate Test Statistic and p-value

$$SE = \sqrt{\frac{(0.789)^2}{48} + \frac{(1.378)^2}{18}}$$

$$SE = \sqrt{\frac{0.622}{48} + \frac{1.898}{18}}$$

$$SE = \sqrt{0.01296 + 0.10544}$$

$$SE = \sqrt{0.11840}$$

$$SE = 0.344$$

Test Statistic

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$t = \frac{(4.125 - 2.611)}{0.344}$$

$$t = 4.40$$

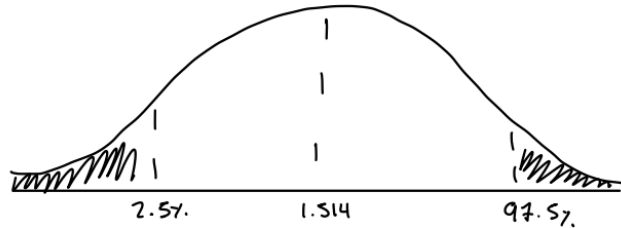
Degree of Freedom ≈ 21

↳ found using Welch's equation

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

$$p\text{-value} = t.cdf(4.40, inf, 21)$$

$$P\text{-value} \approx 0.00024$$



Confidence Interval

$$C = 0.95$$

$$t^* = 2.078$$

$$CI = \bar{x} \pm z \frac{s}{\sqrt{n}}$$

Margin of Error

$$ME = 2.078 \times 0.344$$

$$ME = 0.715$$

Confidence Interval:

$$1.514 \pm 0.715$$

$$(0.799, 2.229)$$